Mental Health Prediction Using Human Emotions With Swin Transformer

A project report submitted in partial fulfilment of the requirement for the Award of the Degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

by

Pran Dutt Avasthi (160722733065)

Moluguru Ruchika (160722733105)

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Under the Guidance of

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Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001.

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We, **Pran Dutt Avasthi (160722733065), Moluguru Ruchika (160722733105)** and **Vikas (160722733085)** students of Methodist College of Engineering and Technology, pursuing Bachelor's degree in Computer Science and Engineering, hereby declare that this major Project report entitled **"Mental Health Prediction Using Human Emotions With Swin Transformer",** carried out under the guidance of **Dr.K.Prasanna** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Computer Science and Engineering. This is a record work carried out by us and the results embodied in this report have not been reproduced/copied from any source.

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INTERNAL **EXTERNAL** **HOD**

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COMPUTER SCIENCE & ENGINEERING DEPARTMENT

Vision & Mission

VISION

To become a leader in providing Computer Science & Engineering education with emphasis on knowledge and innovation.

MISSION

M1: To offer flexible programs of study with collaborations to suit industry needs

M2: To provide quality education and training through novel pedagogical practices

M3: To Expedite high performance of excellence in teaching, research and innovations.

M4: To impart moral, ethical valued education with social responsibility.

Program Educational Objectives

Graduates of Compute Science and Engineering at Methodist College of Engineering and Technology will be able to:

PEO1: Apply technical concepts, Analyse, synthesize data to Design and create novel products and solutions for the real-life problems.

PEO2: Apply the knowledge of Computer Science Engineering to pursue higher education with due consideration to environment and society.

PEO3: Promote collaborative learning and spirit of team work through multidisciplinary projects

PEO4: Engage in life-long learning and develop entrepreneurial skills.

Program Specific Outcomes

At the end of 4 years, Compute Science and Engineering graduates at MCET will be able to:

PSO1: Apply the knowledge of Computer Science and Engineering in various domains like networking and data mining to manage projects in multidisciplinary environments.

PSO2: Develop software applications with open-ended programming environments.

PSO3: Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms

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PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Mental health issues among students have become a growing concern due to academic pressure, social expectations, and personal challenges. Many students experience emotional distress but often fail to recognize early signs of declining mental well-being. To address this problem, this project proposes a system titled “Mental Health Prediction Using Human Emotions with Swin Transformer.” The system analyzes students’ facial expressions to estimate their mental health condition and provide a simple self-assessment score.

The proposed system uses a deep learning approach based on the Swin Transformer model to detect facial emotions from real-time video input. The model identifies seven human emotions: happy, sad, angry, disgust, fear, surprise, and neutral. These emotions are analyzed continuously for 30 seconds using a webcam. Based on the detected emotional patterns, the system calculates a mental health score ranging from 0 to 100, where the scoring begins from a midpoint value of 50. Positive emotional expressions reduce the score slightly, while negative emotional expressions increase it, helping to estimate the emotional state of the student.

The final score provides students with an indication of their current mental health condition. A lower score indicates relatively stable emotional well-being, while a higher score suggests that the student may be experiencing emotional difficulties and may benefit from guidance or support. After the evaluation is completed, the system displays the mental health score and provides a restart option that allows the user to repeat the assessment process.

This project aims to offer a quick, non-invasive, and technology-driven approach for early awareness of mental health conditions among students. By utilizing facial expression recognition and advanced deep learning models, the system helps students understand their emotional state and encourages them to seek timely support when needed.

The Outcome of the project is a Technology based solution which is an Application for **Mental Health Prediction Using Human Emotions With Swin Transformer**.The SDG Indicators addressed by the project are 3.4.1, 3.4.2, 4.1.1, 4.a.1, 9.5.1, 10.2.1

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1. INTRODUCTION

Mental health has become an important concern among students due to academic pressure, social expectations, and personal challenges. Many students experience emotional stress, sadness, frustration, or loss of motivation, which can negatively affect their academic performance and overall well-being. Early identification of emotional changes can help in providing timely support and guidance. With the advancement of Artificial Intelligence and deep learning, it has become possible to analyze human facial expressions to understand emotional states and estimate mental well-being.

Facial Expression Recognition (FER) is a computer vision technique that analyzes facial features to identify human emotions such as happy, sad, angry, disgust, fear, surprise, and neutral. These emotions can provide valuable insights into a person's psychological condition. By observing the patterns of facial expressions over a short period of time, intelligent systems can estimate a person's emotional stability and overall mental health condition in a non-intrusive manner.

The proposed project titled "Mental Health Prediction Using Human Emotions with Swin Transformer" aims to monitor and evaluate the emotional state of students through facial expression analysis. The system captures facial expressions for approximately 30 seconds using a webcam and detects emotions using a deep learning model based on the Swin Transformer architecture. The detected emotions are analyzed to generate a mental health score ranging from 0 to 100, where the scoring begins from a midpoint value of 50. Different emotions influence the score differently; for example, happy expressions may reduce the score slightly while negative emotions may increase it. A higher score indicates a higher risk of poor mental health.

The main objective of this system is to provide self-assessment and early awareness for students regarding their mental health condition. After the analysis, the system displays the calculated mental health score along with an option to restart the process for reassessment. This approach helps students recognize their emotional state and encourages them to seek guidance or support if required. The proposed method uses advanced deep learning techniques such as Swin Transformer models combined with Facial Expression Recognition algorithms, enabling accurate emotion detection and efficient mental health prediction.

1.1 SWIN TRANSFORMER FOR FACIAL EMOTION RECOGNITION

The Swin Transformer is a modern deep learning architecture designed for computer vision tasks. Unlike traditional Convolutional Neural Networks (CNNs), which rely on convolution operations to extract spatial features from images, the Swin Transformer uses a hierarchical transformer mechanism that captures both local and global relationships in an image. This makes it highly effective for facial expression recognition and emotion analysis.

In the proposed system, the Swin Transformer model is used to analyze facial images captured from the webcam in real time. The system processes the video frames of students for approximately 30 seconds, extracts facial features, and predicts the emotional state of the user. The Swin Transformer divides an input image into smaller patches and processes them through self-attention mechanisms within shifted windows. This approach improves efficiency while preserving important facial features such as eye movements, mouth shape, and facial muscle changes.

Mathematically, the self-attention mechanism used in transformer models can be expressed as:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- **Q (Query)** represents the feature vectors used to search for relevant information.
- **K (Key)** represents feature representations of input patches.
- **V (Value)** represents the information associated with the keys.
- $\mathbf{d_k}$ represents the dimensionality of the key vectors.

The Swin Transformer introduces shifted window attention, which allows interaction between neighboring windows in consecutive layers. This improves the model's ability to understand spatial relationships across the entire face.

In this project, the model detects seven facial emotions: Happy, Neutral, Sad, Angry, Fear, Disgust, and Surprise. These emotions are then used to estimate the mental health score of students. The score begins from a midpoint value of 50, and it changes depending on the detected emotional states during the observation period. The final result is displayed as a mental health score out of 100, which helps students perform a quick self-assessment and encourages them to seek guidance if required.

1.2 FACIAL EMOTION RECOGNITION MODEL

Facial Emotion Recognition (FER) is an important application of computer vision that enables machines to detect and interpret human emotions from facial expressions. In this project, FER is implemented using a deep learning model based on the Swin Transformer architecture. The model analyzes facial features from video frames captured through a webcam and classifies them into different emotional categories.

The process begins by capturing real-time video of the student for a duration of 30 seconds. Each frame is processed using computer vision techniques such as face detection and image preprocessing. The detected facial region is then resized and normalized before being passed into the Swin Transformer model.

The deep learning model extracts hierarchical facial features such as:

- Eye movement patterns
- Mouth curvature
- Eyebrow positioning
- Facial muscle tension

These features are used to classify the face into one of the following emotional states:
Happy, Neutral, Sad, Angry, Fear, Disgust, Surprise
After emotion classification, the system evaluates the emotional patterns observed during the time interval and calculates a mental health score. The scoring mechanism is designed specifically for students to help identify early signs of emotional distress.

The scoring process starts from a midpoint score of 50. Based on the detected emotions:

- Positive emotions such as Happy slightly reduce the stress score.
- Neutral emotions maintain the baseline score.
- Negative emotions such as Sad, Angry, Fear, and Disgust gradually increase the score.

The final value is displayed as a Mental Health Score ranging from 0 to 100, where:

- Lower scores indicate better emotional well-being.
- Higher scores indicate possible emotional stress or mental health concerns.

This score provides students with a quick self-assessment tool that can encourage them to seek proper guidance, counseling, or support if their score indicates potential mental health challenges.

1.3 MENTAL HEALTH SCORE CALCULATION

The mental health score calculation module is an important component of the proposed system. Its primary objective is to convert the detected facial emotions into a numerical value that represents the overall emotional well-being of a student.

During the evaluation phase, the system continuously captures facial expressions for 30 seconds. Each frame is analyzed using the trained Swin Transformer model to determine the emotion displayed by the user. The system then calculates a cumulative score based on the detected emotions.

The scoring mechanism is designed with a baseline midpoint of 50, which represents a neutral mental state. Each detected emotion contributes to the final score according to predefined weight values. For example:

- Happy emotion slightly decreases the mental stress score because it indicates positive emotional well-being.
- Neutral emotion does not significantly affect the score.
- Sad, Angry, Fear, and Disgust emotions increase the score since they may indicate emotional distress.

The general scoring formula can be represented as:

$$Mental\ Health\ Score = 50 + \sum_{i=1}^{n} w_i \times E_i$$

Where:

- E_i represents the detected emotion at frame i
- w_i represents the weight assigned to that emotion
- n represents the total number of analyzed frames

After analyzing all frames within the 30-second interval, the system calculates the final score and displays it on the screen.

The output is presented as:

Mental Health Score (0 – 100)

- 0 – 40: Healthy emotional condition
- 41 – 60: Normal emotional state
- 61 – 80: Possible emotional stress
- 81 – 100: High emotional distress

Once the score is displayed, the system also provides a Restart option, which allows the student to repeat the process if they want to reassess their emotional state.

This approach provides a quick and non-invasive mental health monitoring system for students. It encourages early awareness and helps individuals identify emotional difficulties before they develop into more serious mental health problems.

2. LITERATURE REVIEW

2.1 LITERATURE SURVEY

S.No	Year	Title	Method Used	Measures	Key Outcomes	Limitations
1	2025 [1]	Real-Time Classroom Emotion Analysis Using Machine and Deep Learning for Enhanced Student Learning	Hybrid ML + Deep Learning (CNN, SVM, Ensemble models)	Accuracy, real-time performance	Real-time classroom emotion detection improves teaching insights and engagement	Requires high-quality video input; model complexity
2	2024 [2]	Mental Health and Stress Prediction Using NLP and Transformer-Based Techniques	Transformer-based NLP (BERT, RoBERTa)	Accuracy, precision, recall	Transformer models effectively predict stress from text patterns	Limited to text input; contextual ambiguity
3	2023 [3]	Vision Transformer With Attentive Pooling for Robust Facial Expression Recognition	Vision Transformer (ViT) + Attentive Pooling	Accuracy, robustness	Attentive pooling significantly enhances ViT performance for FER	Requires large datasets; computationally heavy
4	2021 [4]	Swin Transformer: Hierarchical Vision Transformer using Shifted Windows	Swin Transformer (Shifted Windows)	Accuracy, speed, scalability	Hierarchical design improves efficiency and accuracy in vision tasks	High computational demand for training
5	2023 [5]	Facial Expression Recognition Algorithm Based on Swin Transformer	Swin Transformer for FER	Accuracy, F1 score	Swin-based FER improves feature extraction and recognition accuracy	Dataset limitations; high model complexity
6	2024 [6]	Swin-FER: Swin Transformer for Facial Expression Recognition	Swin Transformer + optimized FER pipeline	Accuracy, robustness	Achieves high FER accuracy with improved attention modeling	Limited generalization across demographic variations
7	2023 [7]	Facial Expression Recognition using Squeeze and Excitation-powered Swin Transformers	Swin Transformer + SE blocks	Accuracy, feature enhancement	SE blocks further enhance expressive feature extraction	Relatively high computational cost
8	2025 [8]	Mental health assessment model for college students based on facial expression recognition."	CNN + FER-based psychological assessment model	Accuracy, Evaluation metrics	FER-based mental health assessment improves early detection	FER-only; lacks physiological signals

S.No	Year	Title	Method Used	Measures	Key Outcomes	Limitations
9	2025 [9]	Emotional recognition and intervention of students in intelligent English teaching environment based on deep learning	Deep Learning (CNN, RNN, attention models)	Accuracy, intervention effectiveness	Deep learning improves emotion detection in smart English classrooms	Requires continuous monitoring; privacy concerns
10	2025 [10]	Emotion recognition for enhanced learning: using AI to detect students' emotions and adjust teaching methods	AI + FER + adaptive teaching models	Accuracy, adaptability, usability	AI-based emotion detection supports personalized learning	Privacy, need for high-quality video data
11	2025 [11]	Deep learning based emotion recognition for analyzing students' psychological states during competitions	Deep Learning (CNN, Hybrid models)	Accuracy, robustness	Helps assess psychological states during high-pressure events	Limited to competition settings
12	2023 [12]	Real-Time Attention Monitoring System for Classroom: A Deep Learning Approach for Student's Behavior Recognition	Deep Learning-based behavior recognition	Accuracy, response time	Effective real-time monitoring of student attention	Hardware-dependent; requires clear visuals
13	2022 [13]	Real-time mental stress detection using multimodality expressions with a deep learning framework	Multimodal DL (face + voice + physiological signals)	Accuracy, F1 score	High-accuracy stress detection using multimodal fusion	Requires synchronized multimodal inputs
14	2024 [14]	Emotion Recognition of the Elderly Based on Transformer and Physiological Signals	Transformer + physiological features (HRV, ECG, GSR)	Accuracy, sensitivity	Multi-signal fusion improves emotion recognition for elderly users	Elderly-specific dataset; not generalizable

Table 2.1: *Literature Survey*

2.2 GAP ANALYSIS

- Existing mental health detection methods mainly rely on questionnaires or self-reported surveys, which may not always reflect the true emotional condition of students. Many students hesitate to openly share their feelings, resulting in delayed identification of mental health issues such as stress, anxiety, and depression.
- Traditional emotion detection systems often use basic machine learning or simple CNN models, which may fail to capture subtle facial expressions and complex emotional patterns. These approaches sometimes produce less accurate predictions and are not efficient for real-time mental health monitoring.

2.3 OBJECTIVES

- To develop an efficient image preprocessing pipeline including resizing, normalization, and facial feature extraction to improve the performance of the emotion recognition model.
- To design and implement a deep learning model using Swin Transformer to accurately detect human emotions from facial expressions.
- To analyze multiple emotions such as happy, neutral, sad, disgust, angry, fear, and surprise and aggregate them to understand the emotional state of students.
- To calculate a mental health score between 0 and 100 based on the detected emotions, where 50 represents the midpoint of normal mental condition.
- To build a simple interactive system interface that displays the mental health score and allows students to restart the analysis for continuous self-assessment.

2.4 PROBLEM STATEMENT

Rising stress levels are negatively affecting student's emotional well-being and academic performance, yet traditional assessment methods fail to detect early warning signs in real time, making early detection and timely support essential.

3.DESIGN ANALYSIS

3.1 MODEL ARCHITECTURE

This image represents the overall architecture of the Mental Health Prediction System using Human Emotions with Swin Transformer. The system captures facial images of students through a webcam and processes them for emotion detection. Initially, the captured images are resized to 48×48 and converted into grayscale. Preprocessing techniques such as normalization, resizing, and cropping are applied to improve data quality and prepare the images for model input.

The preprocessed images are then passed into a Custom Swin Transformer model, which extracts important facial features. The model processes images in small patches and learns patterns related to different emotional expressions. Some layers remain frozen while others are trained to improve emotion recognition performance.

After feature extraction, a classifier layer categorizes the detected facial expression into seven emotions such as happy, neutral, sad, disgust, angry, fear, and surprise. These emotions are aggregated during the 30-second analysis period to calculate a mental health score between 0 and 100, where 50 represents the normal midpoint. The final result is displayed on the output dashboard, showing the student's mental health score and allowing the user to restart the process for another assessment.

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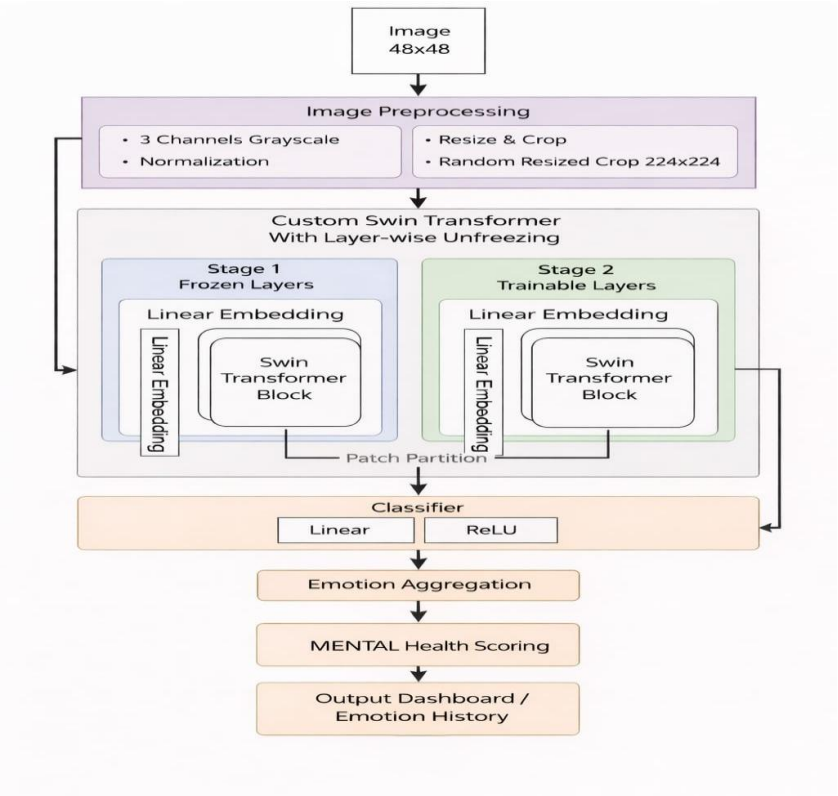


Fig 3.1: *Model Architecture*

3.2 UML DIAGRAMS

Unified Modeling Language (UML) diagrams are a visual representation used in software engineering to model systems and processes. They provide a standardized way to depict the structure and behavior of a system, aiding in communication between stakeholders, designers, and developers . UML diagrams can range from high-level views, such as use case diagrams and activity diagrams, to detailed representations like class diagrams and sequence diagrams. Each type of diagram serves a specific purpose, allowing for clearer understanding, analysis, and design of complex systems. UML diagrams play a crucial role in ensuring software design clarity and efficiency. They assist in defining relationships between different system entities, tracking workflow processes, and identifying dependencies between components. The Use-case diagram helps outline user interactions with the system, ensuring a structured approach to defining functionalities. The Activity Diagram maps the sequence of operations, visualizing the logical flow of user actions and system responses. The Sequence Diagram illustrates object interactions in a time-ordered manner, assisting in refining communication flow between system modules. The Class Diagram provides a detailed representation of object attributes, methods, and relationships, which is essential for implementing robust data structures. As the project evolves, incorporating additional UML diagrams, such as State Diagrams or Component Diagrams, could further enhance understanding and refine system design.

3.2.1 USE CASE DIAGRAM

The use case diagram represents the overall interaction between the users and the Mental Health Prediction System using Human Emotions with Swin Transformer. It explains the workflow of the system from capturing facial images to generating the mental health score. The diagram shows how different actors such as the student, admin, and camera interact with the system to perform various operations.

1. **Start**

The student starts the application through the system interface to begin the mental health assessment process.

2. **Capture Image / Video**

The camera captures the student's facial images or video for a short duration of 30 seconds. This visual input is used by the system to analyze facial expressions.

3. **Upload Image**

The captured frames are automatically uploaded to the system where they are prepared for further processing.

4. **Face Detection**

The system detects the student's face from the captured images. This step ensures that the model

focuses only on the facial region required for emotion detection.

5. **Image Preprocessing**

The detected facial images are preprocessed using techniques such as resizing, normalization, and cropping to improve the quality of the data before feeding it to the model.

6. **Emotion Detection (Swin Transformer)**

The preprocessed images are then passed to the Swin Transformer model, which analyzes facial features and predicts the emotions present in the student's expressions.

7. **Emotion Aggregation**

The detected emotions from multiple frames are aggregated during the 30-second session to determine the overall emotional state of the student.

8. **Mental Health Score Generation**

Based on the aggregated emotions, the system calculates a mental health score between 0 and 100, where 50 represents the midpoint of normal mental condition.

9. **View Score and Emotion History**

Finally, the student can view the mental health score and emotion history on the dashboard, helping them understand their emotional condition and perform self-assessment.

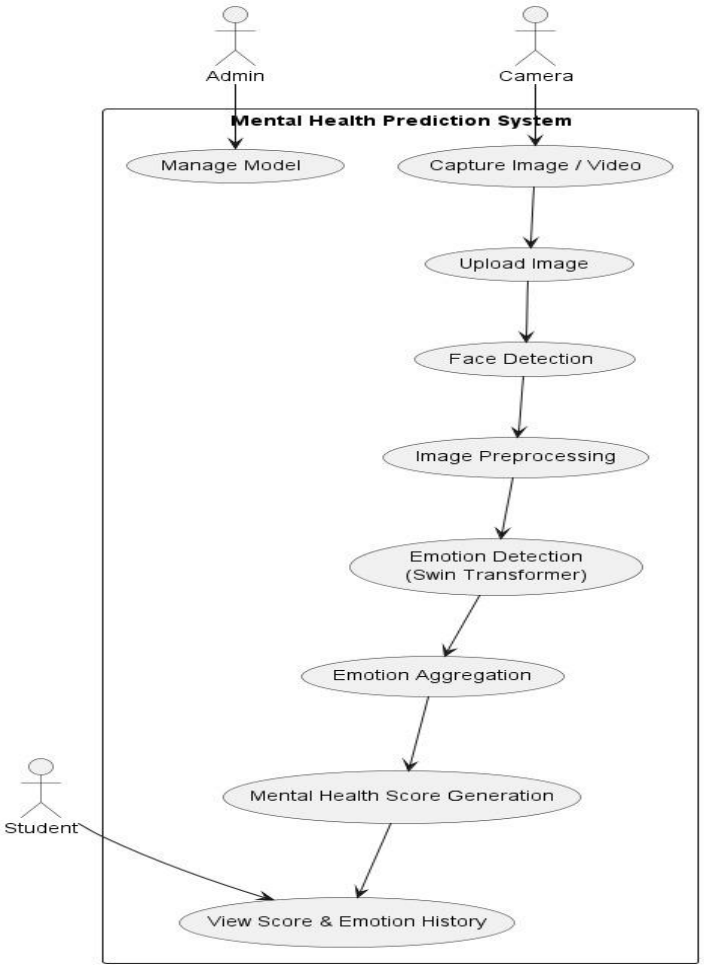


Fig 3.2.1: *Use Case Diagram*

3.2.2 SEQUENCE DIAGRAM

• **Student:** The end-user who interacts with the system by starting the mental health assessment and allowing the camera to capture facial images.
• **User Interface (UI):** The front-end interface responsible for capturing images, sending them to the system, and displaying the mental health score and emotion results.
• **Swin Transformer Model:** The deep learning model used to detect emotions from the captured facial images.
• **Emotion Store:** The component that stores detected emotions during the analysis period.
• **Scoring Engine:** The module responsible for calculating the final mental health score based on detected emotions.

Step-by-Step Description:

1**. Capture / Upload Image**
The student starts the system and the camera captures facial images or video frames during the 30-second analysis period.
2. **Send Image to System**
The captured images are sent from the user interface to the system for processing and analysis.
3. **Emotion Detection**
The system sends the images to the Swin Transformer model, which analyzes facial expressions and detects the emotions present in each frame.
4. **Emotion Result Generation**
The model returns the detected emotion results such as happy, neutral, sad, angry, fear, disgust, and surprise.
5. **Save Emotion Data**
The detected emotions are stored in the Emotion Store, which keeps track of all emotion results collected during the session.
6. **Compute Mental Health Score**
The stored emotions are sent to the Scoring Engine, which calculates the mental health score based on the emotion patterns.
7. **Generate Score Result**
The scoring engine generates a mental health score between 0 and 100, where 50 represents the midpoint of normal mental condition.
8. **Send Score to Interface**
The calculated score and emotion summary are sent back to the user interface.

9. **Display Output**

The system displays the mental health score and emotion history to the student on the dashboard.

10. **Restart Option**

The student can restart the process to perform another mental health assessment if required.

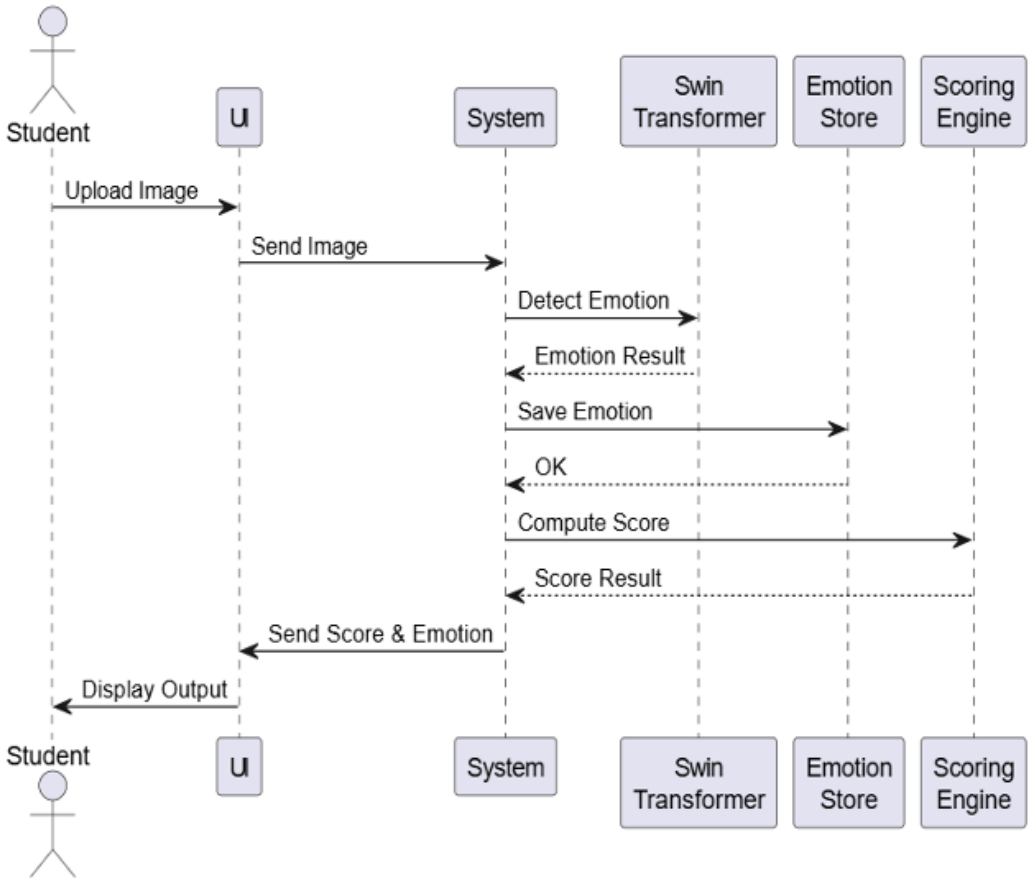


Fig 3.2.2: *Sequence diagram*

3.2.3 ACTIVITY DIAGRAM

The activity diagram represents the overall workflow of the Mental Health Prediction System using Human Emotions with Swin Transformer, describing the process from capturing facial images to displaying the mental health score. It illustrates the sequence of activities performed by the student and the system to analyze emotions and generate the final assessment.

Step-by-Step Description:

1. Start

The student begins the process by opening the application interface and starting the mental health assessment.

2. **Capture Image / Select Video Frame**

The system uses the camera to capture facial images or video frames of the student for approximately 30 seconds, which serve as the input data for emotion analysis.

3. **Store Raw Image**

The captured images are temporarily stored in the system for further processing.

4. **Detect Face**

The system detects the face in the captured images to focus only on the relevant facial region required for emotion recognition.

5. **Image Preprocessing**

The detected facial images undergo preprocessing operations such as cropping, resizing, and normalization to prepare them for the deep learning model.

6. **Run Emotion Detection (Swin Transformer)**

The preprocessed images are passed to the Swin Transformer model, which analyzes facial features and predicts the emotions present in each frame.

7. **Save Emotion Results**

The detected emotions and their confidence values are stored for further analysis during the session.

8. **Aggregate Emotions**

The system aggregates all detected emotions over the 30-second period to determine the overall emotional pattern of the student.

9. **Compute Mental Health Score**

Based on the aggregated emotions, the system calculates a mental health score ranging from 0 to 100, where 50 represents the normal midpoint and higher values may indicate potential mental stress.

10. **Display Results**

Finally, the system displays the mental health score and emotion history on the dashboard, allowing the student to review their emotional state and optionally restart the assessment.

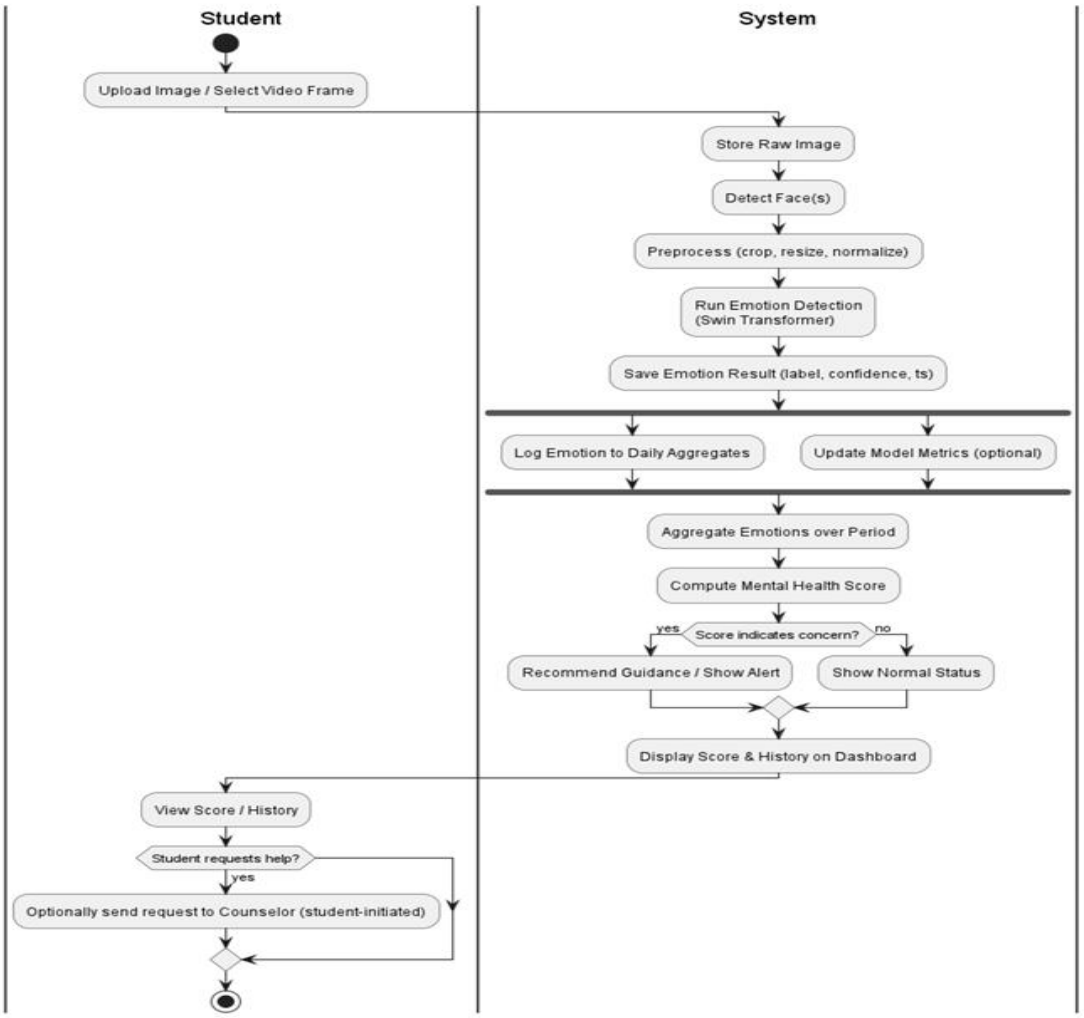


Fig 3.2.3: *Activity diagram*

4.MODULES AND IMPLEMENTATION

4.1 DATA COLLECTION

The dataset used in this project consists of facial expression images and emotion-labelled data used for training and testing the emotion recognition model. Publicly available datasets such as FER2013 and CK+ are commonly used for facial emotion recognition tasks. These datasets contain thousands of grayscale facial images labeled with emotions such as happy, sad, angry, fear, disgust, surprise, and neutral. These images help the model learn pattern in human facial expressions.

```
pip install kaggle

mkdir ~/.kaggle
mv /path/to/kaggle.json ~/.kaggle/
chmod 600 ~/.kaggle/kaggle.json

kaggle competitions download -c challenges-in-representation-learning-facial-expression-recognition-challenge

mkdir datasets
unzip challenges-in-representation-learning-facial-expression-recognition-challenge.zip -d datasets
```

Fig 4.1.: *Dataset loading*

The dataset is downloaded and stored in the project directory using Python tools and libraries. After downloading, the dataset is organized into training and testing folders so that the model can effectively learn from the data and evaluate its performance. Proper dataset organization ensures efficient model training and improves the accuracy of emotion detection.

4.1.1 DATA PRE-PROCESSING

In facial emotion recognition systems, data preprocessing is an important step to improve the performance of the deep learning model. The captured facial images or dataset images are first resized to a standard dimension such as 48×48 pixels, which helps reduce computational complexity and maintain consistency in the input data. The images are then converted into grayscale format, reducing the number of channels and focusing the model on important facial features.

After resizing and conversion, the system performs face detection and cropping to focus on the important facial region. This step removes unnecessary background information and allows the model to concentrate on facial expressions. The images are then normalized by adjusting pixel values using mean and standard deviation, ensuring consistent data distribution across the dataset.

Normalization helps the deep learning model learn more effectively by preventing bias toward

specific images or lighting conditions. These preprocessing steps improve the model's ability to recognize emotional patterns accurately and enhance the overall performance of the Swin Transformer-based emotion detection system used for mental health prediction.

```python
import os
import numpy as np
import pandas as pd
from PIL import Image
from tqdm import tqdm

def preprocess_fer2013(input_csv, output_dir, emotion_labels):
    os.makedirs(output_dir, exist_ok=True)
    for usage in ['train', 'val', 'test']:
        for label in emotion_labels.values():
            os.makedirs(os.path.join(output_dir, usage, label), exist_ok=True)

    df = pd.read_csv(input_csv)
    for index, row in tqdm(df.iterrows(), total=df.shape[0], desc='Processing images'):
        pixels = np.array(row['pixels'].split(), dtype='uint8')
        image = pixels.reshape(48, 48)
        img = Image.fromarray(image)
        label = emotion_labels[row['emotion']]
        usage = row['Usage']
        if usage == 'Training':
            path = os.path.join(output_dir, 'train', label)
        elif usage == 'PublicTest':
            path = os.path.join(output_dir, 'val', label)
        else:
            path = os.path.join(output_dir, 'test', label)
        img.save(os.path.join(path, f'{index}.jpg'))

if __name__ == "__main__":
    emotion_labels = {
        0: 'Angry',
        1: 'Disgust',
        2: 'Fear',
        3: 'Happy',
        4: 'Sad',
        5: 'Surprise',
        6: 'Neutral'
    }
    preprocess_fer2013('fer2013/fer2013.csv', 'FER2013_processed', emotion_labels)
```

Fig 4.1.1: *Data Pre-processing*

4.2 DEEP LEARNING

Deep learning has become an important technology in modern scientific computing and is widely used in many fields such as healthcare, security, image recognition, and emotion analysis. It is a branch of machine learning that focuses on training models using large amounts of data through complex neural network architectures. Deep learning algorithms are capable of automatically learning patterns and features from raw data without the need for manual feature extraction.

Artificial Intelligence (AI) refers to the ability of machines to simulate intelligent human behavior. Machine learning is a subset of AI that enables systems to learn from data and improve their performance over time. Deep learning is a further advancement of machine learning that uses deep neural networks with multiple layers to analyze complex data and make accurate predictions.

Neural networks are composed of multiple layers of interconnected nodes that work similarly to neurons in the human brain. Each node receives input data, processes it using mathematical operations, and passes the result to the next layer. The connections between nodes are assigned weights, which determine the importance of each input in the learning process. During training, these weights are adjusted to improve the model's accuracy.

In this project, deep learning is used to analyze facial expressions and detect emotions from captured images. A Swin Transformer-based deep learning model is used to extract meaningful features from facial images and classify them into emotions such as happy, sad, angry, fear, disgust, surprise, and neutral. These detected emotions are then used to calculate a mental health score for students, helping them understand their emotional condition. Deep learning models require large datasets and powerful computational resources, but they provide highly accurate results for complex tasks such as emotion recognition and mental health prediction.

4.3 NEURAL NETWORKS

A neural network is a computational model inspired by the structure and functioning of the human brain. It is made up of artificial neurons, also known as nodes, that are connected in different layers to process information. Neural networks are widely used in deep learning applications such as image recognition, speech processing, and emotion detection. In this project, neural networks help analyze facial images and identify emotional expressions of students.

A typical neural network consists of three main layers:

- Input Layer
- Hidden Layer(s)
- Output Layer

The input layer receives the input data, which in this project consists of preprocessed facial images captured from the camera. These images contain important visual features such as facial expressions and movements. The input data is then passed to the hidden layers for further processing.

The hidden layers perform complex computations to extract meaningful patterns from the input data. Each neuron in these layers processes the input using mathematical operations involving weights and biases. The connections between neurons determine how information flows through the network and how important each feature is for the prediction.

The output layer produces the final result of the neural network. In this project, the output layer classifies the detected facial expression into one of the seven emotions such as happy, sad, angry, fear, disgust, surprise, and neutral. These predicted emotions are then used by the system to calculate a mental health score for students, helping them understand their emotional state.

Each neuron in the neural network performs several operations during processing:

• The input values are multiplied by their corresponding weights.

• The weighted inputs are summed together to form a weighted sum.

• A bias value is added to adjust the output of the neuron.

• The final value is passed through an activation function, which determines the neuron's output and helps the network learn complex patterns in facial expressions.

4.4 SWIN TRANSFORMER AND EMOTION CLASSIFICATION

4.4.1 SWIN TRANSFORMER

Swin Transformer is a modern deep learning architecture designed for computer vision tasks such as image classification, object detection, and facial emotion recognition. Unlike traditional Convolutional Neural Networks (CNNs), the Swin Transformer processes images using a hierarchical transformer structure and self-attention mechanisms. This allows the model to capture both local and global relationships within an image, making it highly effective for analyzing complex facial expressions.

In the Swin Transformer model, an input image is first divided into small patches, which are then converted into feature embeddings. These embeddings are processed through multiple transformer blocks that use a mechanism called Shifted Window Multi-Head Self-Attention (SW-MSA). This technique allows the model to focus on different regions of the image and learn important facial features such as eyes, eyebrows, and mouth movements that indicate human emotions.

Mathematically, the self-attention mechanism can be represented as:

$$\mathbf{Attention(Q,K,V) = Softmax((QK^T / \sqrt{d_k}))V}$$

Fig 4.4.1: *Self-Attention Mechanism*

Here, Q represents queries, K represents keys, and V represents values derived from the input features. The attention mechanism calculates the relationship between different image patches and helps the model identify important facial regions.

The Swin Transformer introduces a shifted window approach where the attention window moves between layers, enabling cross-window connections. This improves feature learning and enhances the model's ability to recognize subtle facial expressions that are essential for emotion detection.

4.4.2 EMOTION CLASSIFICATION

Emotion classification is the process of identifying human emotions from facial expressions using deep learning models. In this project, the Swin Transformer model is trained on facial emotion datasets such as FER2013, which contain labeled images representing emotions like happy, sad, angry, fear, disgust, surprise, and neutral.

The model extracts important facial features and maps them into emotion categories using a classification layer with the Softmax activation function. The softmax function converts the model outputs into probabilities for each emotion class, and the class with the highest probability is selected as the predicted emotion.

The softmax function is defined as:

$$\mathbf{Softmax}(z_i) = e^{z_i} / \Sigma\, e^{z_j}$$

Fig 4.4.2: *Softmax Classification Function*

During the prediction process, the model analyzes facial expressions captured through the camera and classifies them into one of the seven emotion categories. These detected emotions are then aggregated over a short time period to determine the overall emotional state of the student.

Based on the aggregated emotion results, the system calculates a mental health score ranging from 0 to 100, where 50 represents a neutral baseline and higher values may indicate emotional distress. This approach enables the system to provide early insights into students' mental well-being and support self-assessment through emotion analysis.

4.5 MODEL TRAINING

This step focuses on training a deep learning model to detect and classify human emotions from facial expressions using image data. The training process enables the system to learn patterns from facial images and accurately predict emotions, which are later used to calculate the mental health score of students.

Model Architecture: The model used in this project is based on a Swin Transformer architecture, implemented using the PyTorch deep learning framework. The Swin Transformer serves as the backbone of the model and is responsible for extracting important visual features from facial images. The model uses a pretrained Swin Transformer (swin_base_patch4_window7_224) which has already learned general visual patterns from large image datasets, allowing it to perform more effectively on the facial emotion recognition task.

The input images are first resized to 224 × 224 pixels and normalized using standard mean and standard deviation values. These preprocessing steps ensure that the images are compatible with the Swin Transformer architecture and improve training stability.

```
import torch
from torch.utils.data import DataLoader
from torchvision import transforms, datasets
import timm
from tqdm import tqdm

class CustomSwinTransformer(torch.nn.Module):
    def __init__(self, pretrained=True, num_classes=7):
        super(CustomSwinTransformer, self).__init__()
        self.backbone = timm.create_model('swin_base_patch4_window7_224', pretrained=pretrained, num_classes=0)
        self.classifier = torch.nn.Sequential(
            torch.nn.Linear(self.backbone.num_features, 512),
            torch.nn.ReLU(),
            torch.nn.Dropout(p=0.6),
            torch.nn.Linear(512, num_classes)
        )
```

Fig 4.5(a): *Model Architecture*

The Swin Transformer backbone extracts hierarchical visual features from the facial images. These extracted features are then passed to a custom classification head consisting of fully connected layers. The classifier includes a Linear layer, ReLU activation function, Dropout layer, and a final Linear layer that outputs predictions for the seven emotion classes.

The ReLU activation function introduces non-linearity into the model, allowing it to learn complex facial patterns. The Dropout layer helps prevent overfitting by randomly disabling some neurons during training, which improves the model's generalization ability.

The final classification layer outputs probabilities corresponding to the seven emotion categories: angry, disgust, fear, happy, sad, surprise, and neutral.

Loss Function: During training, the model uses the Cross-Entropy Loss function, which is commonly used for multi-class classification problems. This loss function measures the difference between the predicted emotion probabilities and the actual emotion labels in the dataset.

```python
    def forward(self, x):
        x = self.backbone(x)
        return self.classifier(x)

def train_one_epoch(model, dataloader, optimizer, criterion, device):
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    for inputs, labels in tqdm(dataloader, desc="Training"):
        inputs, labels = inputs.to(device), labels.to(device)
        optimizer.zero_grad()

        outputs = model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item() * inputs.size(0)
        _, preds = torch.max(outputs, 1)
        correct += torch.sum(preds == labels.data)
        total += labels.size(0)

    epoch_loss = running_loss / total
    epoch_acc = correct.double() / total
    return epoch_loss, epoch_acc.item()
```

Fig 4.5(b): *Loss Function*

The model parameters are optimized using the AdamW optimizer, which updates the network weights based on the computed gradients. Training is performed for multiple epochs using batches of training images. During each epoch, the model processes the images, calculates the loss, updates the weights, and computes the training accuracy.

This training process enables the model to learn meaningful facial expression features and accurately classify emotions. Once trained, the model can analyze real-time facial expressions captured from the camera and predict emotions, which are then used by the system to generate the mental health score for students.

```python
if __name__ == "__main__":
    transform = transforms.Compose([
        transforms.Resize((224, 224)),
        transforms.ToTensor(),
        transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
    ])

    train_dataset = datasets.ImageFolder(root="FER2013_processed/train", transform=transform)
    train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)

    model = CustomSwinTransformer(pretrained=True, num_classes=7).to('cuda')
    criterion = torch.nn.CrossEntropyLoss()
    optimizer = torch.optim.AdamW(model.parameters(), lr=1e-4)

    for epoch in range(5):
        loss, acc = train_one_epoch(model, train_loader, optimizer, criterion, 'cuda')
        print(f"Epoch {epoch + 1}: Loss={loss:.4f}, Accuracy={acc:.4f}")
```

Fig 4.5(c): *Dataset Loading and Training Configuration*

Before training, the input images are resized to 224×224 pixels and normalized using predefined mean and standard deviation values. The FER2013 dataset is loaded using the ImageFolder class and processed using DataLoader with batch training. The model is trained for multiple epochs to improve accuracy in emotion classification.

5. TESTING

5.1 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, subassemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the. Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

5.2 TYPES OF TESTING

5.2.1 UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

5.2.2 INTEGRATION TESTING

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components. Test Results: All the test cases mentioned above passed successfully. No defects encountered.

5.2.3 FUNCTIONAL TESTING

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

• Functional testing is centered on the following items

• Valid Input: identified classes of valid input must be accepted.

• Invalid Input: identified classes of invalid input must be rejected.

• Functions: identified functions must be exercised.

• Output: identified classes of application outputs must be exercised.

• Systems/Procedure: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified, and the effective value of current tests is determined.

5.2.4 SYSTEM TESTING

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

5.2.5ACCEPTANCE TESTING

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6. RESULTS

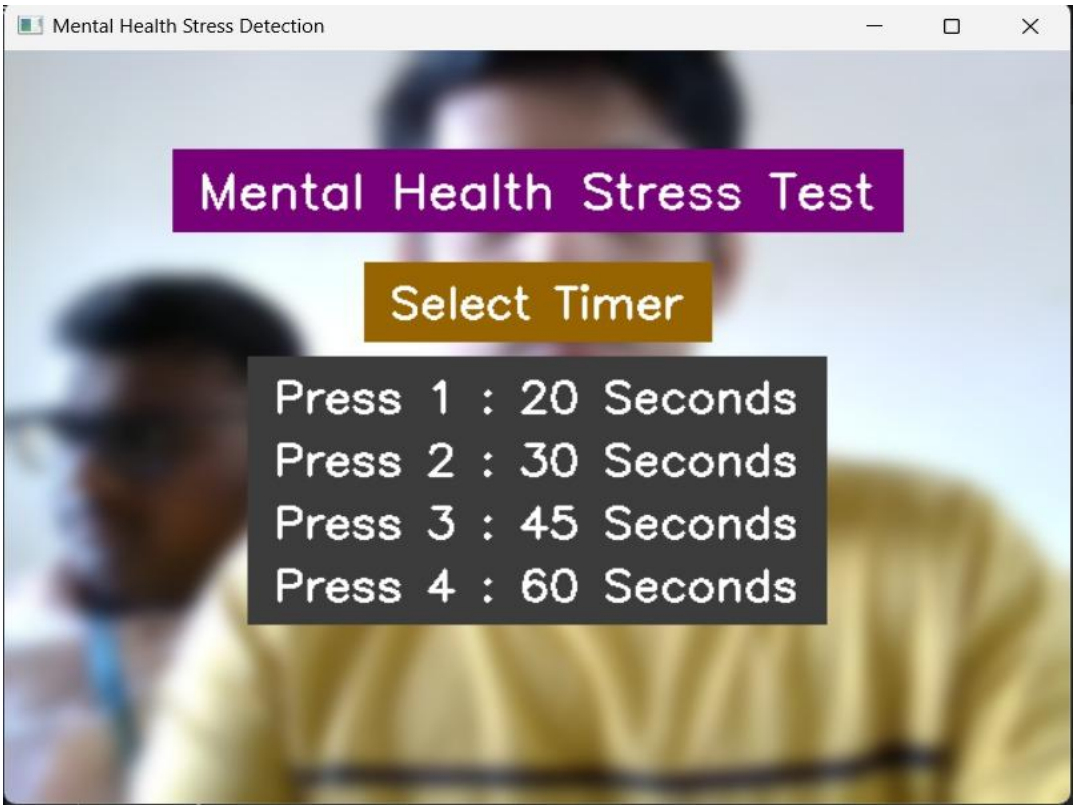
OUTPUT:

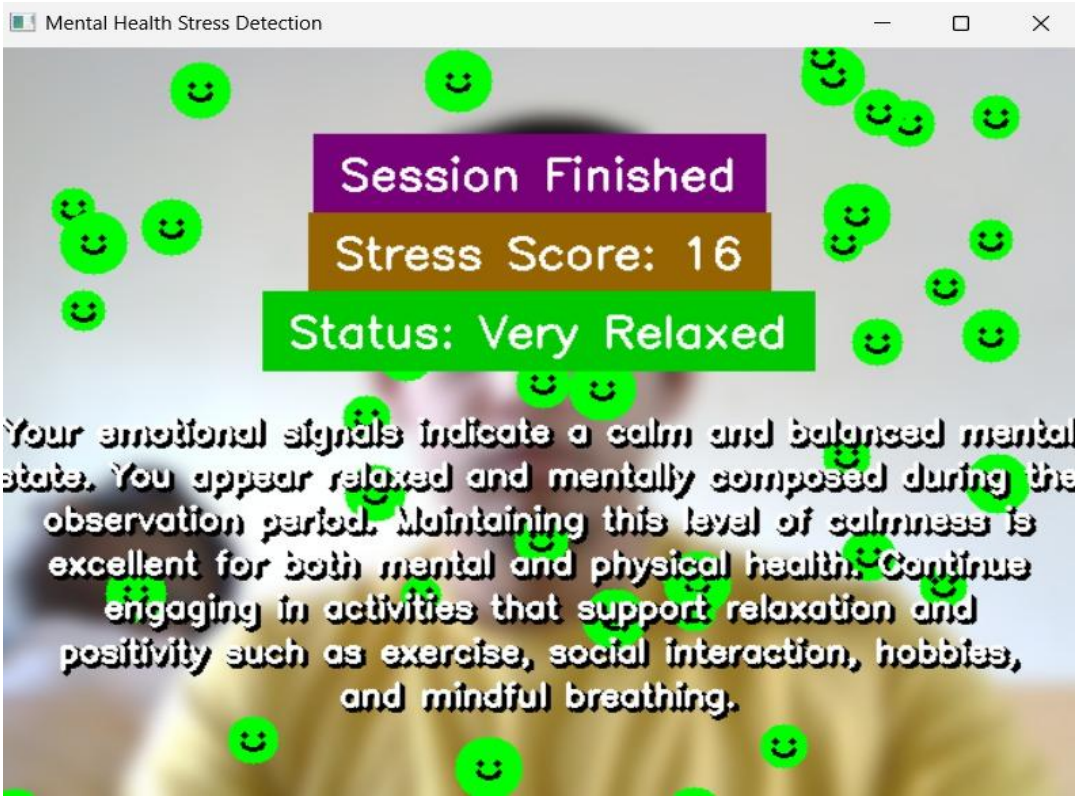
```
100%|██████████| 718/718 [04:52<00:00,  2.46it/s]

Epoch 1
Train Loss: 1.5549  Acc: 0.4652
Val Loss: 1.3653  Acc: 0.5196
Best model saved
100%|██████████| 718/718 [04:50<00:00,  2.47it/s]

Epoch 2
Train Loss: 1.3630  Acc: 0.5195
Val Loss: 1.3023  Acc: 0.5386
Best model saved
100%|██████████| 718/718 [13:34<00:00,  1.14s/it]

Epoch 3
Train Loss: 1.2027  Acc: 0.5585
Val Loss: 1.0338  Acc: 0.5945
Best model saved
100%|██████████| 718/718 [13:36<00:00,  1.14s/it]

Epoch 4
Train Loss: 0.9375  Acc: 0.6448
Val Loss: 1.0046  Acc: 0.6495
Best model saved
100%|██████████| 718/718 [13:34<00:00,  1.13s/it]

Epoch 5
Train Loss: 0.7678  Acc: 0.7015
Val Loss: 0.9269  Acc: 0.6670
Best model saved
100%|██████████| 718/718 [13:34<00:00,  1.13s/it]
```

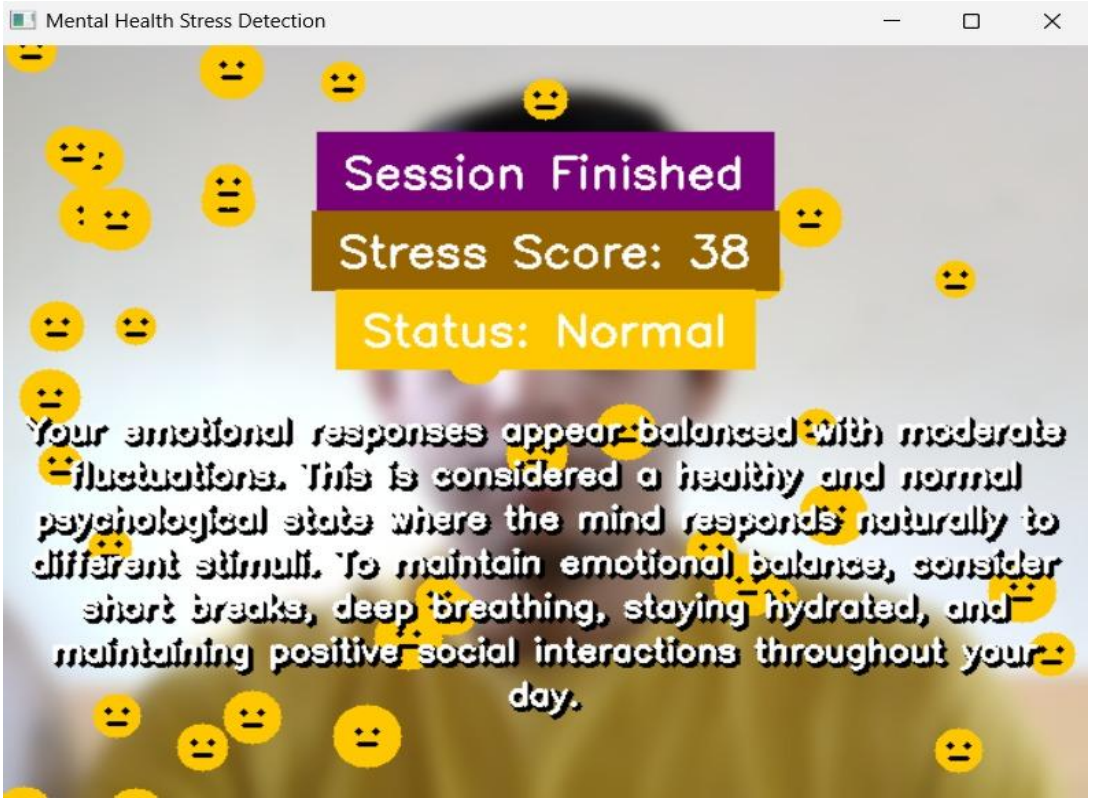
```
Epoch 6
Train Loss: 0.6115  Acc: 0.7627
Val Loss: 1.0059  Acc: 0.6791
Best model saved
100%|██████████| 718/718 [13:34<00:00,  1.13s/it]

Epoch 7
Train Loss: 0.4844  Acc: 0.8140
Val Loss: 1.0225  Acc: 0.6755
100%|██████████| 718/718 [13:34<00:00,  1.13s/it]

Epoch 8
Train Loss: 0.3398  Acc: 0.8725
Val Loss: 1.2262  Acc: 0.6893
Best model saved
100%|██████████| 718/718 [13:35<00:00,  1.14s/it]

Epoch 9
Train Loss: 0.2263  Acc: 0.9165
Val Loss: 1.3822  Acc: 0.6840
100%|██████████| 718/718 [13:34<00:00,  1.13s/it]

Epoch 10
Train Loss: 0.1534  Acc: 0.9447
Val Loss: 1.5512  Acc: 0.6865
100%|██████████| 718/718 [13:34<00:00,  1.13s/it]

Epoch 11
Train Loss: 0.1005  Acc: 0.9628
Val Loss: 1.7732  Acc: 0.6863
Early stopping triggered
Training completed
```

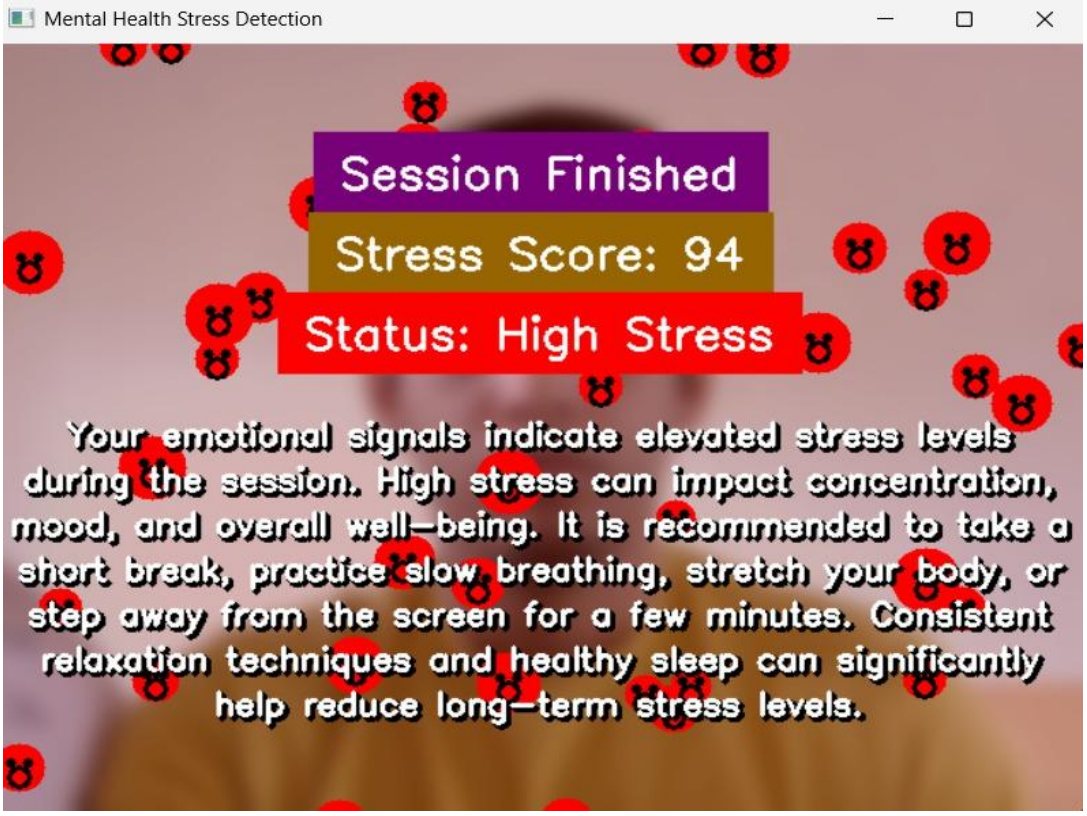
```
100%|██████████| 113/113 [00:52<00:00,  2.14it/s]
Classification Report:
```

	precision	recall	f1-score	support
Angry	0.64	0.58	0.61	491
Disgust	0.67	0.76	0.71	55
Fear	0.55	0.53	0.54	528
Happy	0.92	0.86	0.89	879
Neutral	0.63	0.73	0.68	626
Sad	0.56	0.53	0.54	594
Surprise	0.74	0.83	0.79	416
accuracy			0.69	3589
macro avg	0.67	0.69	0.68	3589
weighted avg	0.69	0.69	0.69	3589


Mental Health Stress Detection
Mental Health Stress Test
Select Timer
Press 1 : 20 Seconds
Press 2 : 30 Seconds
Press 3 : 45 Seconds
Press 4 : 60 Seconds


Mental Health Stress Detection
Session Finished
Stress Score: 16
Status: Very Relaxed
Your emotional signals indicate a calm and balanced mental state. You appear relaxed and mentally composed during the observation period. Maintaining this level of calmness is excellent for both mental and physical health. Continue engaging in activities that support relaxation and positivity such as exercise, social interaction, hobbies, and mindful breathing.


Mental Health Stress Detection
Session Finished
Stress Score: 38
Status: Normal
Your emotional responses appear balanced with moderate fluctuations. This is considered a healthy and normal psychological state where the mind responds naturally to different stimuli. To maintain emotional balance, consider short breaks, deep breathing, staying hydrated, and maintaining positive social interactions throughout your day.


Mental Health Stress Detection
Session Finished
Stress Score: 94
Status: High Stress
Your emotional signals indicate elevated stress levels during the session. High stress can impact concentration, mood, and overall well-being. It is recommended to take a short break, practice slow breathing, stretch your body, or step away from the screen for a few minutes. Consistent relaxation techniques and healthy sleep can significantly help reduce long-term stress levels.

7. CONCLUSION

This project demonstrates an effective approach to predicting student mental health by utilizing facial emotion recognition with deep learning techniques. By analyzing facial expressions captured through images, the system is able to identify emotional patterns that reflect the psychological state of students without requiring manual input or questionnaires. The use of advanced deep learning architecture such as the Swin Transformer enables the model to extract meaningful visual features from facial images and accurately classify emotions including happy, sad, angry, fear, disgust, surprise, and neutral. These detected emotions are then used to generate a mental health score, providing a simple method for students to understand their emotional condition. The system performs real-time analysis within a short time period and presents the results through an interactive interface, allowing users to easily view their score and emotional history. Overall, this project highlights the potential of artificial intelligence and computer vision technologies in supporting mental health awareness among students, offering a non-invasive and efficient approach for early emotional assessment and encouraging timely support or guidance when needed.

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APPENDIX

A. SAMPLE CODE

Model Training

```
import torch
from torch.utils.data import DataLoader, random_split
from torchvision import transforms, datasets
import timm
import numpy as np
from tqdm import tqdm
from sklearn.utils.class_weight import compute_class_weight

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

class CustomSwinTransformer(torch.nn.Module):
    def __init__(self, pretrained=True, num_classes=7):
        super().__init__()

        self.backbone = timm.create_model(
            "swin_base_patch4_window7_224",
            pretrained=pretrained,
            num_classes=0
        )

        self.classifier = torch.nn.Sequential(
            torch.nn.Linear(self.backbone.num_features, 512),
            torch.nn.ReLU(),
            torch.nn.Dropout(0.4),   # reduced dropout
            torch.nn.Linear(512, num_classes)
        )

    def forward(self, x):
        x = self.backbone(x)
        return self.classifier(x)
```

```
train_transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(15),
    transforms.RandomResizedCrop(224, scale=(0.85,1.0)),
    transforms.ColorJitter(
        brightness=0.2,
        contrast=0.2,
        saturation=0.1,
        hue=0.02
    ),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],
                         [0.229,0.224,0.225]),
    transforms.RandomErasing(p=0.25)   # new augmentation
])

val_transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],
                         [0.229,0.224,0.225])
])


dataset = datasets.ImageFolder(
    "/content/pran/FER2013_processed/train",
    transform=train_transform
)
```

```python
val_size = int(0.2 * len(dataset))
train_size = len(dataset) - val_size

train_dataset, val_dataset = random_split(dataset, [train_size, val_size])
val_dataset.dataset.transform = val_transform

train_loader = DataLoader(
    train_dataset,
    batch_size=32,
    shuffle=True,
    num_workers=2,
    pin_memory=True
)

val_loader = DataLoader(
    val_dataset,
    batch_size=32,
    shuffle=False,
    num_workers=2,
    pin_memory=True
)

targets = np.array(dataset.targets)
weights = compute_class_weight(
    class_weight="balanced",
    classes=np.unique(targets),
    y=targets
)
```

```python
weights = torch.tensor(weights, dtype=torch.float).to(device)
criterion = torch.nn.CrossEntropyLoss(weight=weights)


model = CustomSwinTransformer(pretrained=True).to(device)

for param in model.backbone.parameters():
    param.requires_grad = False

optimizer = torch.optim.AdamW(model.parameters(), lr=5e-5)

scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(
    optimizer,
    T_max=20
)

def train_epoch(loader):
    model.train()
    total_loss = 0
    correct = 0
    total = 0

    for x,y in tqdm(loader):
        x,y = x.to(device), y.to(device)

        optimizer.zero_grad()
        out = model(x)
        loss = criterion(out,y)
        loss.backward()
        optimizer.step()
```

```python
        total_loss += loss.item()*x.size(0)
        _,pred = torch.max(out,1)
        correct += (pred==y).sum().item()
        total += y.size(0)

    return total_loss/total, correct/total


def validate(loader):
    model.eval()
    total_loss = 0
    correct = 0
    total = 0

    with torch.no_grad():
        for x,y in loader:
            x,y = x.to(device), y.to(device)
            out = model(x)
            loss = criterion(out,y)

            total_loss += loss.item()*x.size(0)
            _,pred = torch.max(out,1)
            correct += (pred==y).sum().item()
            total += y.size(0)

    return total_loss/total, correct/total
```

```python
best_val_acc = 0
patience = 3
trigger = 0

epochs = 20

for epoch in range(epochs):
    if epoch == 2:
        for param in model.backbone.parameters():
            param.requires_grad = True

    train_loss, train_acc = train_epoch(train_loader)
    val_loss, val_acc = validate(val_loader)

    scheduler.step()

    print(f"\nEpoch {epoch+1}")
    print(f"Train Loss: {train_loss:.4f}  Acc: {train_acc:.4f}")
    print(f"Val Loss: {val_loss:.4f}  Acc: {val_acc:.4f}")

    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), "best_model.pth")
        print("Best model saved")
        trigger = 0
    else:
        trigger += 1
        if trigger >= patience:
            print("Early stopping triggered")
            break

print("Training completed")
```

Model integration in VS Code :

```
import torch
import cv2
import timm
import numpy as np
from torchvision import transforms
import time
import textwrap
import random

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

class_names = ['Angry','Disgust','Fear','Happy','Neutral','Sad','Surprise']

emotion_score_map = {
    'Happy': -2,
    'Neutral': 0,
    'Surprise': -1,
    'Sad': 2,
    'Angry': 2,
    'Fear': 2,
    'Disgust': 2
}
```

```
quotes = {

"Very Relaxed":
"""Your emotional signals indicate a calm and balanced mental state.
You appear relaxed and mentally composed during the observation period.
Maintaining this level of calmness is excellent for both mental and physical health.
Continue engaging in activities that support relaxation and positivity such as
exercise, social interaction, hobbies, and mindful breathing.""" ,

"Normal":
"""Your emotional responses appear balanced with moderate fluctuations.
This is considered a healthy and normal psychological state where the mind
responds naturally to different stimuli. To maintain emotional balance,
consider short breaks, deep breathing, staying hydrated, and maintaining
positive social interactions throughout your day.""" ,

"High Stress":
"""Your emotional signals indicate elevated stress levels during the session.
High stress can impact concentration, mood, and overall well-being.
It is recommended to take a short break, practice slow breathing,
stretch your body, or step away from the screen for a few minutes.
Consistent relaxation techniques and healthy sleep can significantly help
reduce long-term stress levels."""
}

stress_score = 50
duration = None
start_time = None

session_finished = False
menu_screen = True
```

```python
def create_particles(count,width,height):

    particles=[]

    for i in range(count):

        particles.append({

            "x":random.randint(0,width),
            "y":random.randint(0,height),
            "dx":random.uniform(-3,3),
            "dy":random.uniform(-3,3),
            "size":random.randint(12,20)

        })

    return particles

def draw_happy(frame,x,y,size):

    cv2.circle(frame,(x,y),size,(0,255,0),-1)
    cv2.circle(frame,(x-5,y-3),2,(0,0,0),-1)
    cv2.circle(frame,(x+5,y-3),2,(0,0,0),-1)
    cv2.ellipse(frame,(x,y+2),(6,4),0,0,180,(0,0,0),2)


def draw_neutral(frame,x,y,size):

    cv2.circle(frame,(x,y),size,(0,200,255),-1)
    cv2.circle(frame,(x-5,y-3),2,(0,0,0),-1)
    cv2.circle(frame,(x+5,y-3),2,(0,0,0),-1)
    cv2.line(frame,(x-5,y+5),(x+5,y+5),(0,0,0),2)


def draw_scary(frame,x,y,size):

    cv2.circle(frame,(x,y),size,(0,0,255),-1)
    cv2.circle(frame,(x-5,y-3),3,(0,0,0),-1)
    cv2.circle(frame,(x+5,y-3),3,(0,0,0),-1)
    cv2.ellipse(frame,(x,y+6),(6,6),0,0,360,(0,0,0),2)
```

```python
def draw_particles(frame,particles,mode):

    h,w,_ = frame.shape

    for p in particles:

        p["x"] += p["dx"]
        p["y"] += p["dy"]

        if p["x"]<0 or p["x"]>w:
            p["dx"] *= -1
        if p["y"]<0 or p["y"]>h:
            p["dy"] *= -1

        if mode=="happy":
            draw_happy(frame,int(p["x"]),int(p["y"]),p["size"])

        elif mode=="neutral":
            draw_neutral(frame,int(p["x"]),int(p["y"]),p["size"])

        else:
            draw_scary(frame,int(p["x"]),int(p["y"]),p["size"])

def draw_box_text(img,text,y,box_color,scale=0.9):

    (w,h),_=cv2.getTextSize(text,cv2.FONT_HERSHEY_SIMPLEX,scale,2)

    x=(img.shape[1]-w)//2

    cv2.rectangle(img,(x-15,y-h-15),(x+w+15,y+15),box_color,-1)

    cv2.putText(img,text,(x,y),
                cv2.FONT_HERSHEY_SIMPLEX,
                scale,(255,255,255),2)
```

```python
def draw_multiline_text_center(img,text,y):

    lines=textwrap.wrap(text,width=60)

    for i,line in enumerate(lines):

        (w,h),_=cv2.getTextSize(line,cv2.FONT_HERSHEY_SIMPLEX,0.65,2)

        x=(img.shape[1]-w)//2
        y_line=y+(i*28)

        cv2.putText(img,line,(x+2,y_line+2),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    0.65,(0,0,0),4)

        cv2.putText(img,line,(x,y_line),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    0.65,(255,255,255),2)

class CustomSwinTransformer(torch.nn.Module):
    def __init__(self,num_classes=7):

        super().__init__()

        self.backbone=timm.create_model(
            'swin_base_patch4_window7_224',
            pretrained=False,
            num_classes=0
        )

        self.classifier=torch.nn.Sequential(
            torch.nn.Linear(self.backbone.num_features,512),
            torch.nn.ReLU(),
            torch.nn.Dropout(0.6),
            torch.nn.Linear(512,num_classes)
        )
```

```python
    def forward(self,x):

        x=self.backbone(x)

        return self.classifier(x)

model=CustomSwinTransformer()
model.load_state_dict(torch.load("best_model (4).pth",map_location=device))
model.to(device)
model.eval()

transform=transforms.Compose([
    transforms.ToPILImage(),
    transforms.Resize((224,224)),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485,0.456,0.406],
        std=[0.229,0.224,0.225]
    )
])

face_cascade=cv2.CascadeClassifier(
    cv2.data.haarcascades+"haarcascade_frontalface_default.xml"
)

cap=cv2.VideoCapture(0)

particles=create_particles(100,1280,720)

while True:

    ret,frame=cap.read()
    if not ret:
        break

    gray=cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY)
    faces=face_cascade.detectMultiScale(gray,1.3,5)
```

```python
if menu_screen:
    blurred=cv2.GaussianBlur(frame,(35,35),0)
    draw_box_text(blurred,"Mental Health Stress Test",100,(120,0,120),1.0)
    draw_box_text(blurred,"Select Timer",170,(0,100,150))

    draw_box_text(blurred,"Press 1 : 20 Seconds",230,(60,60,60))
    draw_box_text(blurred,"Press 2 : 30 Seconds",270,(60,60,60))
    draw_box_text(blurred,"Press 3 : 45 Seconds",310,(60,60,60))
    draw_box_text(blurred,"Press 4 : 60 Seconds",350,(60,60,60))
    frame=blurred
elif not session_finished:

    if start_time is None:
        start_time=time.time()

    elapsed_time=int(time.time()-start_time)
    remaining_time=max(0,duration-elapsed_time)

    if elapsed_time>=duration:
        session_finished=True
    for (x,y,w,h) in faces:

        face=frame[y:y+h,x:x+w]
        rgb=cv2.cvtColor(face,cv2.COLOR_BGR2RGB)

        input_tensor=transform(rgb).unsqueeze(0).to(device)

        with torch.no_grad():
            output=model(input_tensor)
            _,pred=torch.max(output,1)

            emotion=class_names[pred.item()]

        stress_score+=emotion_score_map[emotion]
        stress_score=max(0,min(100,stress_score))

        cv2.rectangle(frame,(x,y),(x+w,y+h),(255,255,0),2)

        cv2.putText(frame,f"{emotion}",
                    (x,y-10),
                    cv2.FONT_HERSHEY_SIMPLEX,
                    0.8,(255,255,0),2)

    cv2.rectangle(frame,(10,10),(350,90),(0,0,0),-1)
```

```python
    cv2.putText(frame,f"Time Remaining: {remaining_time}s",
                (20,40),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.7,(255,255,255),2)

    cv2.putText(frame,f"Stress Score: {stress_score}",
                (20,75),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.7,(255,255,255),2)


if session_finished:
    blurred=cv2.GaussianBlur(frame,(35,35),0)

    if stress_score<=30:
        status="Very Relaxed"
        mode="happy"
        color=(0,200,0)

    elif stress_score<=60:
        status="Normal"
        mode="neutral"
        color=(0,200,255)

    else:
        status="High Stress"
        mode="scary"
        color=(0,0,255)

        red_overlay=np.full(frame.shape,(0,0,120),dtype=np.uint8)
        blurred=cv2.addWeighted(blurred,0.7,red_overlay,0.3,0)

    draw_particles(blurred,particles,mode)

    draw_box_text(blurred,"Session Finished",90,(120,0,120))
    draw_box_text(blurred,f"Stress Score: {stress_score}",140,(0,100,150))
    draw_box_text(blurred,f"Status: {status}",190,color)

    draw_multiline_text_center(blurred,quotes[status],250)

    draw_box_text(blurred,"Press R to Restart",520,(120,120,0))

    frame=blurred
cv2.imshow("Mental Health Stress Detection",frame)
```

```python
key=cv2.waitKey(1)&0xFF

if key==ord('q'):
    break

if menu_screen:

    if key==ord('1'):
        duration=20
    elif key==ord('2'):
        duration=30
    elif key==ord('3'):
        duration=45
    elif key==ord('4'):
        duration=60
    else:
        duration=None

    if duration:
        menu_screen=False
        start_time=None
        stress_score=50
        session_finished=False

if key==ord('r') and session_finished:

    menu_screen=True
    start_time=None
    stress_score=50
    session_finished=False
    duration=None

cap.release()
cv2.destroyAllWindows()
```

B. SOFTWARE AND HARDWARE REQUIREMENTS

SOFTWARE REQUIREMENTS:

- **Operating System :** Windows 10 or above, Linux
- **Technology :** Python, Deep Learning
- **IDE :** VS Code / Jupyter Notebook
- **Python Version :** 3.10 or above
- **Libraries :** PyTorch, OpenCV, NumPy, Torchvision, Timm

HARDWARE REQUIREMENTS:

- **CPU :** AMD Ryzen 5 7535HS with Radeon Graphics
- **GPU :** NVIDIA GeForce RTX 2050
- **Speed :** 3.30 GHz
- **RAM :** 16 GB

C. TECHNOLOGY USED

PROGRAMMING LANGUAGE

Python: Python, developed by Guido van Rossum in 1991, is a high-level programming language known for its simplicity, readability, and flexibility. Its clear syntax and easy learning curve make it one of the most widely used programming languages in the world. Python supports multiple programming paradigms such as object-oriented, procedural, and functional programming, making it suitable for building a wide range of applications. Due to its strong community support and extensive collection of libraries, Python has become a preferred language for fields such as data science, artificial intelligence, machine learning, and computer vision.

Python plays a crucial role in the development of modern intelligent systems. In the field of machine learning and deep learning, it provides powerful frameworks and libraries that simplify the development of complex models. Libraries such as NumPy, Pandas, and Matplotlib are widely used for data manipulation, numerical computation, and visualization. For building machine learning models, developers commonly use Scikit-learn, which provides tools for classification, regression, clustering, and data preprocessing.

For deep learning applications, Python offers advanced frameworks such as PyTorch, TensorFlow, and Keras, which allow developers to design, train, and deploy neural network models efficiently. In this project, the PyTorch framework is used to implement the Swin Transformer model for facial emotion recognition. Libraries such as OpenCV are used for image processing and face detection, while Torchvision and Timm provide pre-trained models and utilities for building computer vision systems.

Python also integrates well with development environments such as Jupyter Notebook and Visual Studio Code, which provide interactive environments for testing models, visualizing results, and debugging code. Cloud-based platforms like Google Colab and Kaggle further enhance accessibility by allowing developers to train machine learning models using powerful GPUs without requiring expensive hardware.

Due to its versatility, large community support, and powerful ecosystem of libraries, Python continues to be the preferred language for developing intelligent systems and machine learning applications. Its flexibility and scalability make it an ideal choice for implementing advanced technologies such as facial emotion recognition and mental health prediction systems, enabling researchers and developers to build innovative solutions efficiently.

LIBRARIES AND FRAMEWORK

PyTorch: PyTorch is an open-source deep learning framework widely used for building and training neural network models. It provides powerful tools for tensor computation, automatic differentiation, and GPU acceleration. In this project, PyTorch is used to implement and train the Swin Transformer model for facial emotion recognition and mental health prediction.

NumPy: NumPy is a fundamental Python library used for numerical computing and data manipulation. It provides support for multi-dimensional arrays and mathematical operations that are essential for scientific computing. In this project, NumPy helps in handling numerical data and performing efficient computations during image processing and model training.

OpenCV: OpenCV (Open Source Computer Vision Library) is widely used for image and video processing. It provides tools for tasks such as image capturing, face detection, and preprocessing. In this project, OpenCV is used to capture facial images from the camera and prepare them for emotion detection.

Torchvision: Torchvision is a library built on top of PyTorch that provides datasets, model architectures, and image transformation utilities. It is commonly used in computer vision tasks.
In this project, Torchvision is used for image preprocessing operations such as resizing, normalization, and dataset loading.

Timm (PyTorch Image Models): Timm is a popular PyTorch library that provides access to a large collection of pretrained image models. It includes advanced architectures such as Vision Transformers and Swin Transformers. In this project, the Swin Transformer model is loaded using the Timm library to perform facial emotion recognition.

OPTIMIZATION

ADAMW (Adaptive Moment Estimation with Weight Decay):

AdamW is an optimization algorithm widely used in deep learning for training neural networks. It is an improved version of the Adam optimizer that introduces weight decay to improve model generalization and prevent overfitting. AdamW updates the weights of the neural network iteratively based on the training data and helps the model converge faster during the learning process.

The algorithm maintains two moving averages for each parameter: the first moment, which represents the mean of the gradients, and the second moment, which represents the variance of the gradients. These moving averages help stabilize the training process and allow the optimizer to adjust the learning rate adaptively for each parameter.

Compared to traditional stochastic gradient descent, AdamW provides better performance for complex deep learning models because it combines adaptive learning rates with momentum-based updates. The weight decay mechanism also helps reduce overfitting by preventing the model weights from growing too large.

In this project, the AdamW optimizer is used to train the Swin Transformer model for facial emotion recognition. It updates the model parameters during training to minimize the classification loss and improve the accuracy of predicting emotions such as happy, sad, angry, fear, disgust, surprise, and neutral, which are later used to calculate the mental health score of students.

QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)

Course Name:	PROJECT WORK – II	**AY**	2025-26
Course Code:	3PW868CS	**Semester**	VIII
Name of the Guide:	Dr.K Prasanna	**Class**	CSE-B

Mental Health Prediction Using Human Emotions With Swin Transformer

STUDENTS

S. No	HT No	Name of the student
1.	160722733065	Pran Dutt Avasthi
2.	160722733105	Moluguru Ruchika
3.	160722733085	Vikas

Course Outcomes: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	PO
3PW868CS.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	**Apply**	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.2	Analyze and formulate the problem, and design solution of the selected project	**Analyze**	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.3	Evaluate different solutions based on economic and technical feasibility.	**Evaluate**	PO3, PO4, PO5, PO6, PO7, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.4	Effectively plan, develop and deploy a project and confidently perform all aspects of project management	**Create**	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3
3PW868CS.5	Demonstrate effective written and oral communication skills.	**Create**	PO2, PO5, PO7, PO9, PO10, PO12, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
3PW868CS.1	3	2	2	-	-	3	-	2	-	-	2	-	3	3	3
3PW868CS.2	3	3	2	3	2	2	-	2	-	-	2	-	3	3	3
3PW868CS.3	-	-	3	3	2	2	2	2	-	-	3	-	3	2	2
3PW868CS.4	-	-	3	3	3	2	2	3	3	3	3	2	3	3	3
3PW868CS.5	-	2	-	-	2	-	2	2	3	3	2	2	2	2	2

CO-PO-PSO MAPPING (With Performance Indicators)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS01	PSO 2	PSO 3
3PW868CS.1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2	-	-	6.1.1 6.2.1 6.3.1 6.3.2 6.4.1	-	8.2.1 8.2.2 8.2.3	-	-	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	-	**High**	**High**	**High**
3PW868CS.2	1.1.2 1.2.1 1.3.1 1.4.1	2.1.1 2.1.2 2.1.3 2.2.3 2.3.1 2.4.1 2.4.3 2.4.4	3.1.1 3.1.3 3.1.4 3.1.5 3.1.6 3.2.1 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.1 4.1.2 4.1.3 4.2.1 4.3.1 4.3.2 4.3.3 4.3.4	-	6.1.1 6.3.1 6.3.2 6.4.2	-	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	-	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**High**	**High**
3PW868CS.3	-	2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**Mod**	**Mod**
3PW868CS.4	-	2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**High**	**High**	**High**
3PW868CS.5	-	2.2.1 2.2.2 2.2.3 2.2.4	-	-	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	-	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**Mod**	**Mod**	**Mod**

Project Specific Learning Outcomes

LO #	Outcome Description
1	Demonstrate the ability to integrate deep learning architectures such as Swin Transformer to solve real-world challenges in facial emotion recognition for mental health prediction.
2	Analyze facial expressions using computer vision techniques and design an end-to-end system capable of detecting multiple human emotions from image data.
3	Evaluate the performance of the emotion recognition model in terms of prediction accuracy, reliability, and suitability for real-time mental health assessment.
4	Design and implement a complete emotion recognition pipeline, including data preprocessing, model training, testing, and generation of a mental health score based on detected emotions.
5	Demonstrate effective written and oral communication skills in documenting and presenting the project on mental health prediction using facial emotion recognition with deep learning techniques.

PO-WK Mapping:

PO#	PO Description	Mapped WKs	PO#		Mapped WKs
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Environment and sustainability	WK5, WK7
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Ethics	WK7, WK9
PO3	Design / Development of Solutions	WK1, WK2, WK3, WK4, WK5, WK6, WK7	PO9	Individual & Team Work	WK9
PO4	Conduct Investigations of complex problems	WK2, WK3, WK4, WK8	PO10	Communication	WK7, WK9
PO5	Engineering Tool usage	WK2, WK6	PO11	Project Management & Finance	WK5, WK6, WK7
PO6	The Engineer and society	WK1, WK5, WK7	PO12	Life-long Learning	WK8

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	3.4.1, 3.4.2	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.5.1	15 LIFE ON LAND	
4 QUALITY EDUCATION	4.1.1, 4.a.1	10 REDUCED INEQUALITIES	10.2.1	16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

QUALITY ANALYSIS

Project collaboration (Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community ✓	Alumni	Regulating agency	Govt.	Other
Details: Utilized FER2013 dataset for training and testing emotion recognition model for mental health prediction.							

Project Achievements:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes

S. No	Description	Details
1.	Prototype development	Developed a working prototype for emotion-based mental health prediction system
2.	Technology based solution developed	Online Application for Mental Health Prediction Using Human Emotions With Swin Transformer
3.	Publication	In Progress
4.	Start-up incubation	-
5.	Competition prizes	-
6.	Community / Industry / HEI service	Supports mental health awareness and early detection using AI-based emotion analysis
7.	SDG Indicators	3.4.1, 3.4.2, 4.1.1, 4.a.1, 9.5.1, 10.2.1
8.	Grants / Financial Aid	-
9.	Multi-disciplinary	Combines Artificial Intelligence, Computer Vision, and Healthcare
10.	Internship / Employment	-
11.	Self-employment skill	PROSPECTIVE: Potential for startup in AI-based healthcare solutions
12.	Use of real-time data	Uses real-time facial emotion data for prediction

Signature of the Guide **Signature of the Project Coordinator**

QR CODE